

Financial Machine Learning — Midterm Practice Set: Model Answers

QRM3007 • Units 1-7 • Prof. Keehong Rhee

Model answers • 48 questions • answer grid, formula sheet and explanations

Answer Key and Explanations

How to use this key. First check your answers with the grid. Then, for every question, read the explanation, including the *Why not the others* part, even when you answered correctly. Each wrong option is a common mistake, and knowing why it is wrong is the fastest way to learn the idea. Each question ends with a short Korean review note (복습 노트).

Quick answer grid

Q	1	2	3	4	5	6	7	8	9	10	11	12
Ans	B	A	D	A	A	B	C	C	B	D	B	B

Q	13	14	15	16	17	18	19	20	21	22	23	24
Ans	A	D	C	D	A	D	C	C	B	B	C	C

Q	25	26	27	28	29	30	31	32	33	34	35	36
Ans	A	A	D	C	B	A	D	C	D	B	B	A

Q	37	38	39	40	41	42	43	44	45	46	47	48
Ans	C	D	D	B	A	B	C	A	D	A	C	D

Answer distribution: A 12 • B 12 • C 12 • D 12

Formula sheet

	Predicted positive	Predicted negative
Actually positive	TP	FN
Actually negative	FP	TN

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN} \quad F_1 = \frac{2PR}{P + R} \quad \text{Accuracy} = \frac{TP + TN}{N}$$

$$\text{Lasso: } \text{SSE} + \lambda \sum_j |\beta_j| \quad \text{Ridge: } \text{SSE} + \lambda \sum_j \beta_j^2 \quad \text{Gini} = 1 - \sum_k p_k^2$$

Part 1 • Foundations and Financial Data (Units 1-2) — Explanations

Q1 • Answer B (U1)

A bank has 50,000 past loans, and each record says whether the borrower repaid. It wants to judge new applicants. This is:

Correct: B) Supervised learning

The key word is *labels*. Each of the 50,000 past loans comes with the correct answer (repaid or defaulted). The model studies the pairs (applicant features → known outcome) and learns a rule it can apply to new applicants. Learning from examples whose answers are known is exactly **supervised learning**. Because the answer is a category, it is more precisely a supervised *classification* task.

Why not the others

- **A)** No learning is needed: A rule written by hand would not adapt to the data. Learning a rule from 50,000 examples is exactly what ML is for.
- **C)** Reinforcement learning: Reinforcement learning needs an agent that takes actions and receives rewards over time. A bank scoring applicants from a fixed history is not doing that.
- **D)** Unsupervised learning: Unsupervised learning is used when there are no answers. Here every record says whether the borrower repaid.

Remember. Labels present → supervised. Ask first: *does every training example carry the right answer?*

복습 노트(한국어). 정답(label)이 붙은 과거 데이터로 규칙을 배우면 지도학습(supervised learning)이다. 결과가 상환/부도라는 범주이므로 정확히는 지도학습 중 분류(classification) 문제다.

Q2 • Answer A (U1)

A marketing team has the spending records of 1 million customers but no labels. It wants to find natural customer groups. This is:

Correct: A) Unsupervised learning

There is plenty of data (1 million customers) but **no answer key**: nobody has told the algorithm which group each customer belongs to. The algorithm has to discover the structure by itself, for example with k-means clustering. Finding hidden groups in unlabelled data is the classic **unsupervised learning** task.

Why not the others

- **B)** Supervised learning: Supervised learning would need a label for each customer (e.g. 'segment A'). No labels exist.
- **C)** Reinforcement learning: There are no actions or rewards here, only static spending records.
- **D)** Regression: Regression predicts a number from labelled examples. The team wants groups, not numbers.

Remember. No labels + 'find groups' → unsupervised (clustering).

복습 노트(한국어). 레이블 없이 데이터 안의 자연스러운 그룹을 찾는 것은 비지도학습(unsupervised learning), 그중에서도 군집화(clustering)다.

Q3 • Answer D (U1)

A trading bot must sell a huge position. It tries many ways of slicing the order and is rewarded when trading costs are low. This is:

Correct: D) Reinforcement learning

The bot **acts** (chooses how to slice the order), observes the **result** (trading cost), and receives a **reward** when the cost is low. Over many trials it learns which actions earn more reward. This act → reward → improve loop is **reinforcement learning**. Optimal order execution is a well-known RL application in finance.

Why not the others

- **A)** Supervised classification: Supervised classification needs a correct label for each example. Here nobody tells the bot the 'right' slice; it learns from rewards.

- **B) Clustering:** Clustering groups similar points. It does not choose actions.
- **C) Unsupervised learning:** Unsupervised learning looks for structure in data. It has no actions and no rewards.

Remember. Agent + actions + rewards → reinforcement learning.

복습 노트(한국어). 행동(action)을 하고 보상(reward)을 받아 더 나은 행동을 배우는 구조는 강화학습(reinforcement learning)이다. 대량 주문 분할 집행이 대표적인 금융 응용이다.

Q4 • Answer A (U1)

Team A predicts next month's return (e.g. +1.8%). Team B predicts whether a loan will default (yes/no). Their tasks are:

Correct: A) A: regression, B: classification

Look at the **type of the output**. Team A's answer is a number on a continuous scale (+1.8%, -0.4%, ...) → **regression**. Team B's answer is one of two categories (default / no default) → **classification**. Both are supervised, because past returns and past defaults are known.

Why not the others

- **B) A: classification, B: regression:** This reverses the two. A return is a number and a default flag is a category.
- **C) Both: regression:** A yes/no outcome is not a continuous number, so Team B is not regression.
- **D) Both: clustering:** Both teams have labels (past returns, past defaults), so neither task is clustering.

Remember. Number → regression. Category → classification.

복습 노트(한국어). 출력이 숫자(수익률)면 회귀(regression), 범주(부도 여부)면 분류(classification)다. 출력의 형태부터 확인한다.

Q5 • Answer A (U1)

Intern Mina's model scores 99% on the training data but only 51% on new data. Her model is:

Correct: A) Overfitting — it memorised the training data

Training accuracy 99% vs. new-data accuracy 51% is a **48-point gap**. On a two-class problem, 51% is close to a coin flip. The model has memorised details and noise of the training set instead of the general pattern. That is **overfitting** (high variance). Remedies: a simpler model, regularisation, more data, and honest validation.

Why not the others

- **B) Perfect — 99% is excellent:** Only performance on unseen data counts. 99% on training data alone means little.
- **C) Underfitting — it is too simple:** Underfitting means the model is too simple and does badly even on the training data. Here training accuracy is 99%.
- **D) Broken — the new data must be wrong:** Blaming the new data is the usual excuse. A big train-test gap points to the model, not the data.

Remember. Good on train, bad on test → overfitting. Bad on both → underfitting.

복습 노트(한국어). 학습 데이터 99%, 새 데이터 51%처럼 격차가 크면 과적합(overfitting)이다. 학습 데이터에서도 성능이 낮으면 과소적합(underfitting)이다.

Q6 • Answer B (U1)

Before testing, Mina must hold out data. Her bank's 10,000 loan applications come from different customers, so the order of the rows carries no information. What is the standard way to make the test set?

Correct: B) Shuffle the rows at random, then hold out about 20% as the test set

A **train/test split** hides part of the data from the model, so you can score it on examples it has never seen. When the rows are **independent** (each loan is a different customer and the order means nothing), the standard recipe is to **shuffle at random first** and then split, for example 80% for training and 20% for testing. Shuffling makes the two parts look alike, so neither gets, say, only the old or only the large loans, and the test score is a fair estimate. For ordinary machine-learning problems this random split is the default (scikit-learn's `train_test_split` shuffles unless told not to).

Why not the others

- **A)** Use all 10,000 rows for both training and testing: Testing on rows the model was trained on measures memory, not prediction — that is how Mina got 99% in Q5.
- **C)** Test only on the rows the model already predicts well: Picking the test rows by the result is cherry-picking. The score would look far better than reality.
- **D)** Skip the test set and report the training accuracy: Training accuracy says nothing about new data; overfitting hides exactly there.

Remember. Independent rows → shuffle at random, then hold out about 20% as the test set. This random split is the standard.

복습 노트(한국어). 행이 서로 독립이면(고객마다 한 행, 순서에 정보 없음) 무작위로 섞은 뒤 약 20%를 테스트로 떼어 두는 것이 표준이다. 섞어야 학습·테스트 집합이 같은 모습이 되어 점수가 공정해진다.

Q7 • Answer C (U1)

Mina's next data set is a daily time series from 2015 to 2024 — the special case. Which split gives an honest test?

Correct: C) Train on 2015–2022, test on 2023–2024

For independent rows a random shuffle is the standard (previous question), but daily financial data are a **time series**: the order of the days matters. In real life you train on the past and trade in the future. An honest test must copy this, so the test period (2023–2024) must come **after** the training period (2015–2022). Shuffling would let the model 'see' days that lie after the test days, which leaks future information.

Why not the others

- **A)** Train on 2023–2024, test on 2015–2022: Training on the future and testing on the past is backwards. You can never do that in live trading.
- **B)** Train and test on 2015–2024: Testing on data the model was trained on measures memory, not prediction.
- **D)** Shuffle all days, then split 80/20: This is the standard recipe for independent rows, but here it mixes future days into training. Returns and volatility are autocorrelated, so the test looks better than reality.

Remember. Independent rows → random split. Time series is the special case: train on the past, test on the future — never shuffle.

복습 노트(한국어). 일반 데이터는 무작위로 섞어 나누는 것이 표준이지만, 금융 시계열은 예외다. 섞지 않고 과거로 학습해 그 이후 구간으로 테스트해야 실제 운용과 같은 조건이 된다.

Q8 • Answer C (U2)

During an earnings week a stock trades four times its usual dollar volume. Compared with a normal week, dollar bars will be:

Correct: C) About four times as many

A dollar bar closes each time a **fixed dollar amount** (say \$10 million) has traded. If a week normally has \$100 million of trading, it produces about 10 bars. In earnings week \$400 million trades, giving about **40 bars, roughly four times as many**. Dollar bars sample *more* when the market is busy and information is arriving, which is why their returns behave better statistically than time bars.

Why not the others

- **A)** Exactly one per day: One bar per day is again a time bar. Dollar bars are not tied to the clock.
- **B)** Fewer: More trading means the dollar threshold is hit more often, so there are more bars, not fewer.
- **D)** The same number: 'The same number' describes time bars (e.g. one bar per day), which ignore activity.

Remember. Dollar bars follow activity: more dollars traded → more bars.

복습 노트(한국어). 달러 바는 정해진 금액이 거래될 때마다 하나씩 만들어진다. 거래대금이 4배면 바도 약 4배가 된다. 시간 바와 달리 시장 활동량을 따라간다.

Q9 • Answer B (U2)

A fund study uses a database that lists only funds still operating today. The past performance will look:

Correct: B) Too good, because failed funds are missing (survivorship bias)

The database keeps only funds **still operating today**. Funds that performed badly usually closed or merged, so they have been silently dropped. Averaging over the survivors leaves out the losers, and past performance looks **better than it really was**. This is **survivorship bias**. The fix is a survivorship-bias-free database that includes dead funds and delisted stocks.

Why not the others

- **A)** Too bad, because good funds are missing: The bias runs the other way. The *bad* funds are missing, not the good ones.
- **C)** Random: The error is systematic (always upward), not random noise.
- **D)** Exactly right: The sample is not representative, so the result cannot be exactly right.

Remember. Missing losers → results look too good → survivorship bias.

복습 노트(한국어). 현재 살아남은 펀드만 보면 청산된 부진 펀드가 빠져 성과가 실제보다 좋아 보인다. 이것이 생존 편향(survivorship bias)이다.

Q10 • Answer D (U2)

A strategy uses a company's annual earnings, published on 15 March, to trade on 1 January of the same year. This is:

Correct: D) Look-ahead bias

On 1 January the earnings number (published 15 March) **did not exist yet**. A backtest that trades on it is using information from the future, which no real trader could have had. This is **look-ahead bias**. The fix is to timestamp each data point by its *release date* (point-in-time data), not by the period it describes.

Why not the others

- **A)** Regularization: Regularisation is a cure for overfitting, not a type of bias.
- **B)** Survivorship bias: Survivorship bias is about which assets are in the sample, not about when information became available.
- **C)** Overfitting: Overfitting is about model complexity. The problem here is the data timing, even with a simple model.

Remember. Using data before its release date → look-ahead bias. Use point-in-time data.

복습 노트(한국어). 발표 전의 정보를 사용한 백테스트는 미래를 엿본 것(look-ahead bias)이다. 데이터는 '발표일' 기준(point-in-time)으로 정렬해야 한다.

Q11 • Answer B (U1)

A bank has no fraud labels at all. It flags transactions that look very different from normal ones. This is:

Correct: B) Unsupervised anomaly detection

There are **no fraud labels**, so the bank cannot train a supervised classifier. Instead the model learns what *normal* transactions look like and flags points that are far from normal. This is **unsupervised anomaly detection**. Typical tools covered later in the course are LOF, One-Class SVM and Isolation Forest.

Why not the others

- **A)** Supervised regression: Regression would need a labelled numeric target. There is no label of any kind here.
- **C)** Triple-barrier labeling: Triple-barrier labelling creates trade labels from price paths. It is not a fraud detector.
- **D)** Reinforcement learning: There are no actions and rewards, only a set of transactions to screen.

Remember. No labels + 'find the odd ones' → unsupervised anomaly detection.

복습 노트(한국어). 사기 레이블이 없으면 정상 거래의 모습을 학습해 그로부터 멀리 떨어진 거래를 찾는다. 비지도 이상탐지(LOF, One-Class SVM, Isolation Forest)다.

Part 2 • Labels and Linear Models (Units 3-4) — Explanations

Q12 • Answer B (U3)

The price path is 101, 102, 103, 104.5. What is the triple-barrier label?

Correct: B) +1 (the profit-take was touched first)

The triple-barrier method is a **race between three barriers**: profit-take (104), stop-loss (96) and time limit (6 days). Whichever is touched **first** decides the label. Day 1: 101, day 2: 102, day 3: 103 (no barrier yet). Day 4: $104.5 \geq 104$, so the **profit-take** barrier is hit first and the label is **+1**.

Why not the others

- **A)** No label yet: The label is known as soon as any barrier is touched. Day 4 settles it.
- **C)** -1: -1 would require the price to fall to 96 or below before reaching 104. It never went below 101.
- **D)** 0: 0 is given only when the time limit arrives with neither price barrier touched. Here a barrier was touched on day 4.

Remember. Profit-take first → +1. Stop-loss first → -1. Time limit first → 0.

복습 노트(한국어). 삼중 배리어는 이익실현·손절·시간한도 중 '먼저 닿은' 배리어가 레이블을 정한다. 4일차 104.5가 이익실현선(104)을 먼저 넘었으므로 +1이다.

Q13 • Answer A (U3)

Another path is 99, 97, 95.5. What is the label?

Correct: A) -1 (the stop-loss was touched first)

Same trade, different path: 99 (day 1), 97 (day 2), 95.5 (day 3). On day 3 the price is $95.5 \leq 96$, so the **stop-loss** barrier is touched first and the label is **-1**. The profit-take barrier at 104 was never close.

Why not the others

- **B)** +1: +1 needs a touch of 104 first. The price only went down.

- C) No label yet: The stop-loss touch on day 3 settles the label immediately.
- D) 0: 0 applies only if no price barrier is touched within 6 days. The stop-loss was hit on day 3.

Remember. Read the path in time order and stop at the first barrier touched.

복습 노트(한국어). 3일차 95.5가 손절선(96) 아래로 내려갔으므로 손절이 먼저다. 레이블은 -1이다.

Q14 • Answer D (U3)

A third path stays between 98 and 103 for all six days. What is the label?

Correct: D) 0 (the time limit was reached first)

The price wanders between 98 and 103 for all 6 days. It never reaches **104** (profit-take) or **96** (stop-loss). The **vertical barrier** (time limit) is therefore reached first, and the label is **0**: no meaningful move happened within the holding period.

Why not the others

- A) No label yet: The time limit always ends the race, so every trade gets a label.
- B) -1: -1 needs a touch of 96. The minimum was 98.
- C) +1: +1 needs a touch of 104. The maximum was 103.

Remember. Neither price barrier within the time limit → label 0.

복습 노트(한국어). 6일 동안 104도 96도 닿지 않았으므로 시간한도(수직 배리어)가 먼저 도달했다. 레이블은 0이다.

Q15 • Answer C (U4)

To fit a straight line through data points, ordinary least squares (OLS) chooses the coefficients that make what as small as possible?

Correct: C) The sum of squared errors

For each data point, the **error** (residual) is the actual value minus the fitted value, $e_i = y_i - \hat{y}_i$. OLS chooses the intercept and slopes that minimise $\sum_i e_i^2$, the **sum of squared errors (SSE)**. Squaring makes positive and negative errors count equally and punishes large misses more.

Why not the others

- A) The number of points: The number of points is fixed by the data. The method cannot change it.
- B) The slope: The slope is an *output* of the fit, not the quantity being minimised.
- D) The largest coefficient: Shrinking coefficients is what Ridge/Lasso *add*. Plain OLS has no penalty on coefficient size.

Remember. OLS: minimise $\sum (y_i - \hat{y}_i)^2$.

복습 노트(한국어). OLS는 잔차 제곱합 $\sum (y_i - \hat{y}_i)^2$ 을 최소화하는 계수를 고른다. 계수 크기에 대한 벌점은 없다.

Q16 • Answer D (U4)

An analyst tests 400 candidate factors and suspects most of them are useless. She wants the model to keep only a short list. Which method fits?

Correct: D) Lasso (L1), which sets weak coefficients exactly to zero

Lasso adds the **L1 penalty** $\lambda \sum_j |\beta_j|$ to the squared-error loss. Because of the corners of the absolute-value function, the best solution often sets many coefficients **exactly to zero**. With 400 mostly useless factors, Lasso keeps a short list and drops the rest. This is automatic **feature selection**. A larger λ gives a shorter list.

Why not the others

- **A)** Adding more factors: Adding factors makes the problem worse: more noise, more overfitting.
- **B)** Plain OLS with all 400 factors: OLS with 400 factors and no penalty will overfit badly and keep every useless factor.
- **C)** Ridge (L2), which keeps every factor: Ridge shrinks coefficients toward zero but almost never exactly to zero, so all 400 factors stay in.

Remember. Lasso (L1) = shrink **and delete** → feature selection.

복습 노트(한국어). Lasso(L1) 벌점 $\lambda \sum |\beta_j|$ 는 약한 계수를 정확히 0으로 만든다. 그래서 변수 선택(feature selection)이 자동으로 이루어진다.

Q17 • Answer A (U4)

6-month and 12-month momentum are almost the same signal. In OLS their coefficients jump to +48 and -47. The analyst wants to keep both but calm them down. Which method fits?

Correct: A) Ridge (L2), which shrinks both without deleting either

When two features are almost identical, OLS cannot tell them apart, so their coefficients become unstable: one goes very positive (+48), the other very negative (-47), and the two nearly cancel. Ridge adds the **L2**

penalty $\lambda \sum \beta_j^2$, which makes such huge coefficients expensive. It **shrinks both smoothly** (e.g. to about +0.6 and +0.5) and shares the effect between them, without deleting either.

Why not the others

- **B)** Plain OLS: Plain OLS is what produced the unstable +48 / -47 in the first place.
- **C)** No method can help: Ridge regularisation is designed for exactly this case (multicollinearity).
- **D)** Lasso (L1), which deletes one of them: Lasso tends to pick one of two correlated features and drop the other, but the analyst wants to keep both.

Remember. Ridge (L2) = shrink **but keep** → stabilises correlated features.

복습 노트(한국어). 상관성이 매우 높은 두 변수는 OLS에서 계수가 +48, -47처럼 불안정해진다. Ridge(L2)는 둘 다 남기면서 크기를 안정적으로 줄인다(수축).

Q18 • Answer D (U4)

A portfolio has groups of correlated factors, and the analyst wants to shrink them and also delete pure noise.

The Elastic Net penalty $\lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$ fits because it:

Correct: D) Combines the L1 (Lasso) and L2 (Ridge) penalties

The Elastic Net penalty has two parts: $\lambda_1 \sum |\beta_j|$ is the **L1 (Lasso)** part, which can delete noise features, and $\lambda_2 \sum \beta_j^2$ is the **L2 (Ridge)** part, which shrinks groups of correlated features together. Using both gets the strengths of each: grouped shrinkage *and* sparsity.

Why not the others

- **A)** Uses only the L2 penalty: L2 alone (Ridge) would never delete the pure-noise factors.
- **B)** Uses only the L1 penalty: L1 alone (Lasso) would arbitrarily drop members of a correlated group.
- **C)** Uses no penalty: With no penalty this is plain OLS, which neither shrinks nor deletes.

Remember. Elastic Net = L1 + L2: delete like Lasso, shrink groups like Ridge.

복습 노트(한국어). Elastic Net은 L1과 L2를 함께 쓴다. 상관된 변수 묶음은 Ridge처럼 함께 수축하고, 잡음 변수는 Lasso처럼 0으로 만든다.

Q19 • Answer C (U4)

Match each method to its penalty: Lasso, Ridge, Elastic Net.

Correct: C) L1, L2, L1 + L2

A summary table to memorise:

Method	Penalty	Formula	Effect
Lasso	L1	$\lambda \sum \beta_j $	shrinks, some to exactly 0
Ridge	L2	$\lambda \sum \beta_j^2$	shrinks, none to 0
Elastic Net	L1 + L2	both terms	shrinks groups and deletes noise

Why not the others

- **A) L1 + L2, L2, L1:** This scrambles all three. Only Elastic Net combines L1 + L2.
- **B) L2, L1, L1 + L2:** This swaps Lasso and Ridge. Lasso uses absolute values (L1); Ridge uses squares (L2).
- **D) L1, L2, no penalty:** Elastic Net does have a penalty, the sum of L1 and L2.

Remember. Lasso ↔ L1 (abs values). Ridge ↔ L2 (squares). Elastic Net ↔ both.

복습 노트(한국어). Lasso = L1(절댓값), Ridge = L2(제곱), Elastic Net = L1 + L2. 짝을 그대로 기억한다.

Part 3 • Precision, Recall and F1 — Three Stories — Explanations

Q20 • Answer C (U5)

Model A labels every passenger 'civilian'. Its accuracy is 99.999%, and the team is proud. What is wrong?

Correct: C) Its recall is 0% — it catches none of the 10 terrorists

Build the confusion matrix for Model A (positive = terrorist): TP = 0, FN = 10, FP = 0, TN = 999,990. Accuracy =

$(0 + 999,990) / 1,000,000 = 99.999\%$, which looks great. But $\text{Recall} = \frac{TP}{TP + FN} = \frac{0}{10} = 0\%$: it catches **no**

terrorists. When the positive class is rare, a model that never says 'positive' still gets very high accuracy.

Accuracy is the wrong KPI here, and recall is the one that measures the mission.

Why not the others

- **A) It flags too many civilians:** Model A flags zero passengers, so it cannot flag too many civilians.
- **B) Nothing — 99.999% accuracy is excellent:** 99.999% accuracy is earned by doing nothing. The security chief's goal (catch terrorists) is completely missed.
- **D) Its precision is too high:** Precision is undefined or meaningless for Model A because it flags nobody (TP + FP = 0). It is not 'too high'.

Remember. Rare target → accuracy misleads. Ask 'how many of the real positives did we catch?' → **recall**.

복습 노트(한국어). 테러리스트가 100만 명 중 10명뿐이면 '전원 민간인' 모델도 정확도 99.999%가 나온다. 하지만 재현율(recall)은 0/10 = 0%다. 희귀한 표적에서는 정확도가 아니라 재현율을 봐야 한다.

Q21 • Answer B (U5)

The chief says: 'Catch every terrorist!' **Model B** labels every passenger 'terrorist'. What is its recall?

Correct: B) 100%

Model B labels **everyone** 'terrorist'. All 10 real terrorists are among the flagged, so TP = 10 and FN = 0.

Recall = $\frac{10}{10+0} = 100\%$. Perfect recall is trivially easy: just flag everybody.

Why not the others

- **A)** 50%: There is no reason for 50%. Every terrorist is flagged.
- **C)** 0%: 0% is Model A's recall, from flagging nobody.
- **D)** 0.001%: 0.001% is Model B's *precision* (10 / 1,000,000), not its recall.

Remember. Flag everyone → recall = 100% automatically. So recall alone can be gamed.

복습 노트(한국어). '전원 테러리스트'로 판정하면 10명을 모두 잡으므로 재현율은 100%다. 즉 재현율 하나만으로는 쉽게 속일 수 있다.

Q22 · Answer B (U5)

Model B also flags 999,990 innocent civilians. Which metric exposes this problem?

Correct: B) Precision — only 10 of 1,000,000 flags are correct (0.001%)

Model B's matrix: TP = 10, FP = 999,990, FN = 0, TN = 0. Precision = $\frac{TP}{TP + FP} = \frac{10}{1,000,000} = 0.001\%$.

Out of every 100,000 people stopped, only one is a real terrorist. Precision asks: **of the people we flagged, how many were right?** That is the metric that exposes the cost of false alarms.

Why not the others

- **A)** No metric can show it: Precision shows the problem clearly.
- **C)** The number of passengers: The passenger count is just the sample size. It does not measure the model's mistakes.
- **D)** Recall — it is 100%: Recall is 100% here. It *hides* the problem, because it ignores false positives.

Remember. Recall can be gamed by flagging everyone; **precision** stops that trick. You need both.

복습 노트(한국어). 전원을 표시하면 정밀도(precision) = 10/1,000,000 = 0.001%다. 정밀도는 '표시한 사람 중 실제로 맞은 비율'이며, 무조건 표시하는 꼼수를 드러낸다.

Q23 · Answer C (U5)

The team needs one score that is high only when both precision and recall are high. Which metric is that?

Correct: C) The F1 score, $F_1 = \frac{2PR}{P+R}$

We want one score that is high **only if both** P and R are high. The F1 score is their **harmonic mean**:

$F_1 = \frac{2PR}{P+R}$. The harmonic mean is pulled toward the *smaller* value. For example, P = 1.0 and R = 0.0 give F1 = 0, while a simple average would give 0.5. So neither Model A (R = 0) nor Model B (P ≈ 0) can score well.

Why not the others

- **A)** The number of flags: The number of flags is not a quality measure at all.
- **B)** Accuracy: Accuracy was already shown to be misleading for rare targets (Model A: 99.999%).
- **D)** Recall alone: Recall alone is gamed by Model B (100% by flagging everyone).

Remember. F1 = harmonic mean of P and R. Low if **either** is low.

복습 노트(한국어). F1 = 2PR/(P+R)은 정밀도와 재현율의 조화평균이다. 둘 중 하나라도 낮으면 F1이 낮아지므로 둘을 함께 달성해야 높은 점수가 나온다.

Q24 • Answer C (U5)

Model C flags 20 passengers and catches 8 of the 10 terrorists. So precision = $8/20 = 0.4$ and recall = $8/10 = 0.8$. What is its F1?

Correct: C) About 0.53

Step by step: $P = 8/20 = 0.4$, $R = 8/10 = 0.8$. $F_1 = \frac{2 \times 0.4 \times 0.8}{0.4 + 0.8} = \frac{0.64}{1.2} \approx 0.533$. Note that F1 (0.53) is below the simple average (0.60) because the harmonic mean leans toward the weaker metric.

Why not the others

- **A) 0.60:** 0.60 is the arithmetic mean $(0.4 + 0.8)/2$. F1 uses the harmonic mean, which is lower.
- **B) 0.40:** 0.40 is the precision alone.
- **D) 0.80:** 0.80 is the recall alone.

Remember. Compute F1 as $2PR/(P+R)$, not $(P+R)/2$.

복습 노트(한국어). $P = 0.4$, $R = 0.8$ 이면 $F1 = 0.64/1.2 \approx 0.53$ 이다. 단순평균 0.60보다 낮다. 조화평균은 작은 값 쪽으로 끌린다.

Q25 • Answer A (U5)

Ranked by F1, which model should the chief choose?

Correct: A) Model C — its F1 (≈ 0.53) is far above A (0) and B (≈ 0)

Summary of the three models:

Model	Precision	Recall	F1
A (all civilian)	—	0%	0
B (all terrorist)	0.001%	100%	≈ 0.00002
C	40%	80%	≈ 0.53

A fails on recall and B fails on precision. Only **Model C** balances the two, so it wins clearly on F1. The story: accuracy → misleading; recall → needed; precision → stops 'flag everyone'; F1 → balances both.

Why not the others

- **B)** They are all equally good: Their F1 scores are 0, ≈ 0 and 0.53, which are clearly not equal.
- **C)** Model A — it has the highest accuracy: Accuracy rewards Model A for catching nobody. That is the trap this story warns about.
- **D)** Model B — it has the highest recall: Model B's 100% recall comes with 0.001% precision, so its F1 is about 0.

Remember. Choose the model by the KPI that matches the mission. With two competing goals, use F1.

복습 노트(한국어). A는 재현율 0, B는 정밀도 ≈ 0 이라 F1이 모두 ≈ 0 이다. 둘의 균형을 잡은 C($F1 \approx 0.53$)를 고른다. 흐름: 정확도의 함정 → 재현율 → 정밀도 → F1.

Q26 • Answer A (U3)

What is the recall (sensitivity) of this test?

Correct: A) 0.90

Recall (also called **sensitivity**) = share of truly infected people that the test catches.

Recall = $\frac{TP}{TP + FN} = \frac{90}{90 + 10} = \frac{90}{100} = 0.90$. Read it from the '**Actually infected**' row of the table.

Why not the others

- **B)** 0.945: 0.945 is the accuracy, $(90 + 855) / 1,000$.
- **C)** 0.10: 0.10 is the miss rate $(FN / 100 = 10 / 100)$, which is $1 - \text{recall}$.
- **D)** 0.67: 0.67 is the precision, $90 / (90 + 45)$.

Remember. Recall reads across the **actual-positive row**: $TP / (TP + FN)$.

복습 노트(한국어). 재현율(민감도) = $TP/(TP+FN) = 90/100 = 0.90$. 혼동행렬의 '실제 감염' 행을 가로로 읽는다.

Q27 • Answer D (U3)

What is the precision of this test?

Correct: D) About 0.67

Precision = share of positive test results that are truly infected.

Precision = $\frac{TP}{TP + FP} = \frac{90}{90 + 45} = \frac{90}{135} \approx 0.667$. Read it down the '**Test positive**' column. About one in three positives is a false alarm.

Why not the others

- **A)** About 0.33: 0.33 is the *false-alarm share* among positives $(45 / 135)$, which is $1 - \text{precision}$.
- **B)** 0.945: 0.945 is the accuracy.
- **C)** 0.90: 0.90 is the recall.

Remember. Precision reads down the **predicted-positive column**: $TP / (TP + FP)$.

복습 노트(한국어). 정밀도 = $TP/(TP+FP) = 90/135 \approx 0.67$. '검사 양성' 열을 세로로 읽는다. 양성 판정의 약 3분의 1이 오경보다.

Q28 • Answer C (U3)

The 10 infected people who tested negative were sent home. In the table they are:

Correct: C) False negatives (FN)

Name each cell by two words. The second word is what the test **said** (positive/negative); the first word says whether that was **true or false**. These 10 people are infected but tested **negative**, and the test was **wrong**, so they are **false negatives (FN)**. In this story they are the dangerous cell: infected people sent home.

Why not the others

- **A)** True positives (TP): True positives are infected people who tested positive (the 90).
- **B)** False positives (FP): A false positive is a *healthy* person who tested positive (the 45).
- **D)** True negatives (TN): True negatives are healthy people who tested negative (the 855).

Remember. Second word = prediction; first word = was it correct? Infected + negative test → FN.

복습 노트(한국어). 감염자인데 음성으로 나온 10명은 위음성(FN)이다. 뒤 단어는 예측(Positive/Negative), 앞 단어는 그 예측이 맞았는지(True/False)를 뜻한다.

Q29 • Answer B (U3)

Given the story (false alarms are cheap, missed cases are dangerous), the clinic should care most about:

Correct: B) High recall — miss as few infected people as possible

Compare the costs. A **false positive (FP)** means one extra PCR test, which is cheap. A **false negative (FN)** means an infected person goes home and infects others, which is costly. Recall = $TP / (TP + FN)$ falls when FN rises, so the clinic should push **recall** as high as possible, even at the price of more false alarms (lower precision).

Why not the others

- **A)** High precision — never raise a false alarm: Precision guards against false alarms, but here false alarms are cheap because the PCR test catches them.
- **C)** High accuracy only: Accuracy is dominated by the 855 healthy negatives and says little about missed infections.
- **D)** Low recall: Lower recall means *more* missed infections, the opposite of the goal.

Remember. FN costly → maximise recall. FP costly → maximise precision.

복습 노트(한국어). 오경보(FP)는 PCR 재검사로 걸러지니 비용이 작고, 놓친 감염자(FN)는 전파 위험이 크다. FN이 비쌀수록 재현율을 우선한다.

Q30 • Answer A (U5)

The risk team asks: 'Of all the frauds, how many did we stop?' What is that number?

Correct: A) Recall = 15 / 20 = 75%

Matrix (positive = fraud): TP = 15 (blocked and really fraud), FP = 50 - 15 = 35 (blocked but honest), FN = 20 - 15 = 5 (fraud that got through), TN = 10,000 - 15 - 35 - 5 = 9,945. 'Of all the frauds, how many did we stop?'

is **recall** = $\frac{15}{15+5} = \frac{15}{20} = 75\%$.

Why not the others

- **B)** Precision = 15 / 50 = 30%: 30% is precision, which answers a different question (of the blocks, how many were right?).
- **C)** 15 / 10,000 = 0.15%: 15 / 10,000 is just the share of all transactions that were caught frauds. It is not a standard metric.
- **D)** Accuracy = 99.6%: Accuracy = (15 + 9,945) / 10,000 = 99.6%. It looks great but is dominated by honest transactions.

Remember. 'Of all the real positives, how many did we catch?' → recall.

복습 노트(한국어). TP 15, FP 35, FN 5, TN 9,945. '전체 사기 중 몇 건을 막았나'는 재현율 = 15/20 = 75%다. 정확도 99.6%는 정상 거래에 묻혀 의미가 작다.

Q31 • Answer D (U5)

The call centre complains: 'Most blocked cards belonged to honest customers!' Which metric measures this complaint?

Correct: D) Precision = 15 / 50 = 30%

Using the same matrix, the complaint is about the **blocks**: of 50 blocked cards, only 15 were fraud and **35**

were honest customers. That is **precision** = $\frac{15}{15+35} = \frac{15}{50} = 30\%$. In other words, 70% of blocks were false alarms. Raising the threshold would usually raise precision but lower recall: this is the precision-recall trade-off.

Why not the others

- **A)** The number of transactions: The transaction count is not a quality measure.
- **B)** Recall = 15 / 20 = 75%: Recall (75%) is about frauds *missed*, not honest customers *blocked*.
- **C)** Accuracy: Accuracy mixes everything together and hides the 35 false blocks among 9,945 correct passes.

Remember. 'Of the ones we flagged, how many were right?' → precision. Complaints about false alarms → precision.

복습 노트(한국어). '차단한 카드 중 실제 사기는 몇 건인가'는 정밀도 = $15/50 = 30\%$ 다. 정상 고객 35명이 차단됐다. 오경보에 대한 불만은 정밀도의 문제다.

Part 4 • Classifiers and Anomaly Detection (Units 5-6) — Explanations

Q32 • Answer C (U5)

A loan officer wants the model to say 'this applicant has a 23% chance of default'. Which model gives a probability like this directly?

Correct: C) Logistic regression

Logistic regression computes a linear score $z = \beta_0 + \beta_1 x_1 + \dots$ and passes it through the **sigmoid**

$\sigma(z) = \frac{1}{1 + e^{-z}}$, which always lies between 0 and 1. The output is read directly as a probability. For example,

$z = -1.2$ gives $\sigma(-1.2) \approx 0.23$, a **23% chance of default**.

Why not the others

- **A)** A dollar bar: A dollar bar is a way to sample market data. It is not a model.
- **B)** K-means clustering: K-means is unsupervised clustering. It assigns groups, not default probabilities.
- **D)** Plain OLS on a 0/1 label: OLS on a 0/1 label can predict values below 0 or above 1 (e.g. -0.2 or 1.3), which are not valid probabilities.

Remember. Need a probability for a yes/no outcome → logistic regression (sigmoid squashes to 0-1).

복습 노트(한국어). 로지스틱 회귀는 선형 점수 z 를 시그모이드 $\sigma(z) = 1 / (1 + e^{-z})$ 에 넣어 0과 1 사이의 확률을 낸다. OLS는 0 미만이나 1 초과 값을 낼 수 있다.

Q33 • Answer D (U5)

Most transactions sit in a dense crowd; one transaction sits far away in empty space. LOF (Local Outlier Factor) will:

Correct: D) Flag it, because its local density is much lower than its neighbours'

LOF compares each point's **local density** with the density of its neighbours. A point inside the dense crowd has about the same density as its neighbours, so $LOF \approx 1$ (normal). The isolated transaction lives in empty space, so its density is much **lower** than that of its nearest neighbours, LOF is much greater than 1, and it is **flagged as an outlier**.

Why not the others

- **A)** Mark the crowd as anomalies: Points in the crowd have $LOF \approx 1$, which is normal. Only the lonely point stands out.
- **B)** Ignore it, because it has no label: LOF is unsupervised. It needs no labels at all.
- **C)** Delete the whole crowd: The dense crowd is the normal behaviour, and LOF keeps it.

Remember. LOF: much lonelier than your neighbours → anomaly.

복습 노트(한국어). LOF는 한 점의 국소 밀도를 이웃의 밀도와 비교한다. 이웃보다 훨씬 외로운 점(LOF가 1보다 훨씬 큼)을 이상치로 표시한다.

Q34 • Answer B (U6)

Repaid customers have credit scores 2, 3, 4 and defaulters have 8, 9. Where does an SVM put the boundary?

Correct: B) At 6, the middle of the widest gap

An SVM picks the boundary with the **widest margin**, the widest empty 'road' between the classes. The closest repaid customer is at 4 and the closest defaulter is at 8, so the gap runs from 4 to 8. The boundary sits in the **middle**: $(4 + 8) / 2 = 6$, giving a margin of 2 on each side.

Why not the others

- **A) At 8:** At 8 the boundary would touch a defaulter: margin 0 on that side.
- **C) At 5:** At 5 the margins are uneven (1 and 3). The SVM maximises the *smaller* side, so it moves to 6.
- **D) At 4:** At 4 the boundary would touch a repaid customer: margin 0 on that side.

Remember. SVM boundary = middle of the widest gap between classes.

복습 노트(한국어). SVM은 두 집단 사이의 가장 넓은 빈 도로(margin) 한가운데에 경계를 둔다. 가장 가까운 두 점 4와 8의 중간인 6이다.

Q35 · Answer B (U6)

In that example, which customers decide where the boundary goes?

Correct: B) Only the closest ones, at 4 and 8 (the support vectors)

Only the points **on the edge of the margin** determine the boundary: the repaid customer at 4 and the defaulter at 8. These are the **support vectors**. Moving the customer at 2 to 1, or the one at 9 to 10, would leave the boundary at 6. Moving the point at 4 would change it.

Why not the others

- **A) All five customers equally:** Points far from the boundary (2, 3, 9) have no influence on an SVM.
- **C) Only the one at 2:** The point at 2 is the farthest from the gap, so it has no influence.
- **D) Only the one at 9:** The point at 9 is behind the closest defaulter (8), so it has no influence.

Remember. Support vectors = closest points to the boundary; they alone fix it.

복습 노트(한국어). 경계를 정하는 것은 마진 가장자리의 점(4와 8), 즉 서포트 벡터(support vector)뿐이다. 멀리 있는 점은 움직여도 경계가 바뀌지 않는다.

Q36 · Answer A (U6)

A new bank has years of normal transactions but no examples of fraud. Which model can still learn a 'fence' around normal behaviour?

Correct: A) One-Class SVM

With **no fraud examples**, a normal two-class classifier cannot be trained. A **One-Class SVM** learns only from normal data: it draws a tight boundary (a 'fence') around where normal transactions live. Anything that falls **outside** the fence is flagged.

Why not the others

- **B) Supervised classification with fraud labels:** The bank has no fraud labels, so a supervised classifier with fraud labels is impossible.
- **C) Ordinary least squares:** OLS predicts a number from labelled data and does not learn a boundary around normal behaviour.
- **D) Logistic regression:** Logistic regression needs examples of both classes (fraud and normal) to fit.

Remember. Only normal data available → One-Class SVM (or another unsupervised detector).

복습 노트(한국어). 사기 사례가 없으면 정상 데이터만으로 울타리를 그리는 One-Class SVM을 쓴다. 울타리 밖의 거래를 이상으로 표시한다.

Part 5 • Trees and Ensembles (Unit 7) — Explanations

Q37 • Answer C (U7)

What does a decision tree do?

Correct: C) Asks a series of yes/no questions (e.g. 'income > \$50k?') and gives the decision at the leaf it reaches

A decision tree is a **flowchart of yes/no questions**. Starting at the **root** (e.g. 'income > \$50k?'), each answer sends the applicant down a branch to the next question ('debt ratio < 40%?'), until a **leaf** is reached. The leaf holds the decision, usually the majority class of training loans that landed there. The tree chooses each question to make the groups as **pure** as possible (lowest Gini impurity).

Why not the others

- **A)** Averages the answers of the nearest applicants: Averaging the nearest applicants describes KNN, not a tree.
- **B)** Groups applicants without using any labels: A tree is trained on labels (repaid/default), so it is supervised, not clustering.
- **D)** Draws one straight line through all applicants: One straight line is a linear model (e.g. logistic regression). Trees split the space into boxes.

Remember. Tree = root question → branches → leaves with decisions. Splits chosen for purity.

복습 노트(한국어). 의사결정 트리는 예/아니오 질문의 흐름도다. 루트에서 시작해 질문을 따라 내려가면 리프(leaf)에 결정이 있다. 각 분할은 그룹이 가장 순수해지도록(지니 불순도 최소) 고른다.

Q38 • Answer D (U7)

The analyst lets the tree keep splitting until every leaf holds a single applicant. Training accuracy is 100%, but accuracy on new applicants falls to 60%. What happened?

Correct: D) Overfitting — the tree memorised the training applicants

If the tree keeps splitting until **every leaf holds one applicant**, each leaf just stores that person's answer. Training accuracy is 100% by construction: it has **memorised** the training set, including noise and one-off cases. New applicants do not match those quirks, so accuracy drops to 60%. The 40-point gap is **overfitting**: a fully grown tree has very high variance.

Why not the others

- **A)** Nothing — 100% training accuracy is the goal: 100% training accuracy is a warning sign, not a goal. Only performance on new data matters.
- **B)** The new applicants' data are wrong: Blaming the data is the usual excuse. A 100% vs. 60% gap points to the model.
- **C)** Underfitting — the tree is too small: An underfitting tree would also do badly on training data. Here training accuracy is perfect.

Remember. A fully grown tree memorises → overfits. It must be stopped early (pruned).

복습 노트(한국어). 리프마다 한 명씩 남을 때까지 쪼개면 학습 데이터를 외운 것이다. 학습 100%, 새 데이터 60%의 격차가 과적합의 신호다. 트리는 성장을 제한해야 한다.

Q39 • Answer D (U7)

To stop this, the analyst sets max_depth = 4. This means:

Correct: D) At most 4 questions from the root down to any leaf

max_depth = the maximum number of questions on any path from the root to a leaf. With **max_depth** = 4, no applicant is asked more than 4 questions, so the tree can have at most $2^4 = 16$ leaves. A shallower tree cannot carve out tiny, noisy regions.

Why not the others

- **A)** Only 4 features may be used: The number of features is controlled by other settings (e.g. **max_features** in a random forest), not **max_depth**.
- **B)** Only 4% of the data is used: **max_depth** says nothing about how much data is used.
- **C)** Exactly 4 trees are trained: The number of trees is **n_estimators** in a forest. A single tree is still one tree.

Remember. **max_depth** caps the **length of each path** → simpler tree, less overfitting.

복습 노트(한국어). **max_depth**는 루트에서 리프까지 거치는 질문 수의 상한이다. 4면 리프가 최대 $2^4 = 16$ 개로 제한된다.

Q40 · Answer B (U7)

*She also sets **min_samples_split** = 50. This means:*

Correct: B) A node is split only if it holds at least 50 applicants

min_samples_split = the smallest number of samples a node must have before the tree is **allowed to split it**. With 50, a node holding 49 or fewer applicants becomes a leaf and is not split further. This stops the tree from splitting small groups to chase a few unusual cases.

Why not the others

- **A)** The tree is split into 50 parts: It does not split the tree into 50 parts.
- **C)** Every split uses 50 features: Each split asks about one feature. This parameter is about sample counts, not features.
- **D)** The data are split into 50 folds: Folds belong to cross-validation, which is a different concept.

Remember. **min_samples_split**: 'Don't split a node smaller than N.'

복습 노트(한국어). **min_samples_split**은 '이 수보다 작은 노드는 더 쪼개지 않는다'는 규칙이다. 소수의 특이 사례를 쫓는 분할을 막는다.

Q41 · Answer A (U7)

*Next she sets **min_samples_leaf** = 30. This means:*

Correct: A) Every leaf must contain at least 30 applicants

min_samples_leaf = every leaf (final group) must contain **at least 30 applicants**. A split that would create a leaf with, say, 3 applicants is not allowed. A rule learned from 3 people is mostly noise; a rule backed by 30+ people is more trustworthy. Difference from **min_samples_split**: *split* checks the parent **before** splitting; *leaf* checks each child **after** splitting.

Why not the others

- **B)** Each leaf uses 30 features: Leaves hold samples, not features.
- **C)** The tree must have exactly 30 leaves: It sets a minimum *size* per leaf, not the *number* of leaves (that is **max_leaf_nodes**).
- **D)** The tree is trained 30 times: A single tree is trained once. The number 30 is about samples per leaf.

Remember. **min_samples_leaf**: 'Every leaf must have at least N samples.'

복습 노트(한국어). min_samples_leaf는 '모든 리프에 최소 이만큼의 표본이 있어야 한다'는 규칙이다. split은 분할 전 부모 노드를, leaf는 분할 후 자식 노드를 검사한다.

Q42 · Answer B (U7)

Finally she sets max_leaf_nodes = 8. This means:

Correct: B) The tree may end in at most 8 leaves (8 final decisions)

max_leaf_nodes = the maximum number of **leaves** (final decision boxes) the tree may have. With 8, the tree grows its best splits until it has 8 leaves and then stops. Fewer leaves → simpler rules → less overfitting, and a tree that is easier to explain to a credit committee.

Control	What it limits	Effect of a smaller / larger value
max_depth	questions per path	smaller → simpler
max_leaf_nodes	total leaves	smaller → simpler
min_samples_split	size needed to split	larger → simpler
min_samples_leaf	size of each leaf	larger → simpler

Why not the others

- **A)** Each leaf holds 8 applicants: That would be min_samples_leaf (and it is a minimum, not an exact count).
- **C)** The tree must use 8 features: It limits leaves, not features.
- **D)** 8 trees vote together: Voting trees belong to a random forest, not to a single tree setting.

Remember. All four controls make the tree smaller; they are the main defences against tree overfitting.

복습 노트(한국어). max_leaf_nodes는 리프(최종 결정 상자)의 총 개수 상한이다. 네 가지 통제 모두 트리를 작게 만들어 과적합을 막는다(깊이·리프 수는 작게, 분할·리프 표본 수는 크게).

Q43 · Answer C (U7)

For a new applicant, 320 trees say 'default' and 180 trees say 'repay'. What does the forest predict?

Correct: C) Default — the majority vote (320 of 500)

For **classification**, each of the 500 trees casts one vote. 320 vote 'default' and 180 vote 'repay'. The forest takes the **majority vote**, so the prediction is **default**. The vote share $320/500 = 0.64$ can also be read as an estimated probability of default. Combining many different trees (each trained on a bootstrap sample with random features) cancels out individual trees' errors and lowers variance.

Why not the others

- **A)** The answer of the first tree only: Using one tree would throw away the benefit of the ensemble.
- **B)** No decision — the trees disagree: Disagreement is normal. Voting exists exactly to resolve it.
- **D)** Repay — the minority wins: The majority decides, not the minority.

Remember. Random forest classification = majority vote. The vote share $\approx$ probability.

복습 노트(한국어). 랜덤 포레스트 분류는 다수결이다. 500그루 중 320그루가 '부도'이므로 예측은 부도이며, 득표율 0.64는 부도 확률 추정치로 읽을 수 있다.

Q44 • Answer A (U7)

The bank also uses a forest to predict the loss amount (a number) if a loan defaults. How does the forest combine its 500 trees?

Correct: A) It averages the 500 trees' predictions

For **regression** (a numeric target such as the loss amount), votes make no sense. Instead the forest **averages**

the trees' numbers: $\hat{y} = \frac{1}{500} \sum_{b=1}^{500} \hat{y}_b$. For example, if the trees predict amounts around \$40k, the forest outputs their mean. Averaging many noisy but different predictions reduces variance.

Why not the others

- **B)** It multiplies the predictions: Multiplying predictions has no statistical meaning here.
- **C)** It uses only the best tree: Picking one tree loses the variance reduction of the ensemble.
- **D)** It takes the largest prediction: Taking the maximum would make every prediction too high, which is biased upward.

Remember. Classification → vote. Regression → average.

복습 노트(한국어). 랜덤 포레스트 회귀는 트리들의 예측값을 평균한다. 서로 다른 트리의 오차가 상쇄되어 분산이 줄어든다. 분류는 투표, 회귀는 평균.

Q45 • Answer D (U7)

A random forest belongs to a family of ensemble methods called **bagging** (bootstrap aggregating). What does bagging do?

Correct: D) Trains many trees side by side, each on a random bootstrap sample of the loans, then combines them by vote or average

Bagging = **bootstrap + aggregating**. Draw many bootstrap samples (random samples of the 2,000 loans, drawn with replacement, each the same size). Train one tree on each sample — independently, so all trees can be trained at the same time (**in parallel**). Then aggregate: majority vote for a class, average for a number. Each tree is noisy in its own way, and combining them cancels much of that noise, so the variance falls. A random forest is bagging plus one extra trick: each split looks at only a random subset of the features, which makes the trees even more different from each other.

Why not the others

- **A)** Trains one very deep tree on all the loans: One deep tree is the overfitting problem (Q38) that bagging is meant to fix.
- **B)** Trains trees one after another, each fixing the previous tree's mistakes: That is boosting (next question).
- **C)** Removes the labels and groups the loans: That is clustering. Bagging is supervised and uses the labels.

Remember. Bagging = many trees in parallel, each on a bootstrap sample → vote or average.

Random forest = bagging + random features.

복습 노트(한국어). 배깅(bootstrap aggregating)은 복원추출한 표본마다 트리를 따로(병렬로) 학습해 투표나 평균으로 합친다. 개별 트리의 잡음이 상쇄되어 분산이 줄어든다. 랜덤 포레스트는 배깅에 무작위 특성 선택을 더한 것이다.

Q46 • Answer A (U7)

Another team uses **boosting** (for example AdaBoost or gradient boosting). How does boosting build its trees?

Correct: A) One after another: each new small tree concentrates on the loans the trees so far got wrong, and the trees are added up

Boosting is **sequential**. Start with a simple, shallow tree (a 'weak learner'). Look at the loans it got wrong and give them more weight (AdaBoost), or fit the next tree to the errors that are left (gradient boosting). Repeat: each new tree **corrects the mistakes** left by the trees before it. The final model is the weighted sum of all the trees. Because each tree depends on the ones before, the trees must be trained in order, not in parallel.

Why not the others

- **B)** All at once, each on a random bootstrap sample, then a majority vote: That is bagging (previous question).
- **C)** It keeps only the single best tree: Boosting combines many trees; one shallow tree on its own is weak.
- **D)** It trains one tree and copies it 500 times: Identical copies add nothing. Each boosted tree differs because it targets the errors that are still left.

Remember. Boosting = small trees in sequence, each fixing the mistakes of the ones before → weighted sum.

복습 노트(한국어). 부스팅은 얇은 트리를 차례로(순차적으로) 쌓는다. 앞 트리가 틀린 사례에 가중치를 더 주거나 남은 오차를 다음 트리가 학습해 실수를 고쳐 나간 뒤 모두 가중합한다. 대표 예: AdaBoost, 그래디언트 부스팅 (XGBoost, LightGBM).

Q47 · Answer C (U7)

Which statement about bagging and boosting is correct?

Correct: C) Both are ensemble methods: bagging trains trees in parallel and mainly reduces variance; boosting trains them in sequence and mainly reduces bias

An **ensemble** combines many models into one prediction. Bagging and boosting are its two main families.

Bagging: independent trees in parallel on bootstrap samples, combined by vote or average; it mainly lowers variance (deep trees that overfit become stable). Example: random forest. **Boosting:** shallow trees in sequence, each fixing the previous errors, combined by a weighted sum; it mainly lowers bias (weak, simple trees become a strong model). Examples: AdaBoost, gradient boosting.

Why not the others

- **A)** Bagging and boosting are unsupervised methods: Both learn from labels (default or repaid), so both are supervised.
- **B)** Boosting trains its trees in parallel; bagging trains them in sequence: This is reversed.
- **D)** Only boosting is an ensemble method: Both are ensembles. A random forest, a bagging method, combines 500 trees.

Remember. Ensemble → bagging (parallel, lowers variance, random forest) or boosting (sequential, lowers bias, AdaBoost / gradient boosting).

복습 노트(한국어). 앙상블은 여러 모델을 합쳐 하나의 예측을 만든다. 그 안에 두 갈래가 있다. 배깅은 병렬·독립 학습으로 분산을, 부스팅은 순차 학습으로 편향을 주로 줄인다.

Q48 · Answer D (U7)

Account balances are 10, 11, 12, 13 and 45 (thousand dollars). Isolation Forest cuts the range at random. Which balance is usually isolated with the fewest cuts?

Correct: D) 45 — it sits alone in empty space

Isolation Forest cuts the range at **random points** until a value sits alone. The gap between 13 and 45 is **32 units**, while the whole range from 10 to 45 is 35, so a random cut lands in that gap about $32/35 \approx 91\%$ of the time. One cut is usually enough to isolate 45. Values in the tight cluster 10–13 need several cuts to be separated from each other. **Short path = anomaly.**

Why not the others

- **A)** 10: 10 is inside the dense cluster; separating it from 11 needs a cut in a 1-unit gap.
- **B)** 12: 12 is in the middle of the cluster and needs the most cuts.
- **C)** All need the same number of cuts: Points in empty space are isolated much faster than points in a crowd. That difference is the whole idea of the method.

Remember. Isolation Forest: anomalies are few and far away → isolated with few cuts.

복습 노트(한국어). Isolation Forest는 무작위로 자르며 점을 고립시킨다. 13과 45 사이의 빈 공간이 넓어 45는 한두 번 만에 고립된다. 경로가 짧을수록 이상치다.